

Ratios, Rates & Data

ANSWER KEY



Ratios and proportions

- 1 Simplify the ratio 18 : 24 to lowest terms.

Divide both terms by 6: 3 : 4.

- 2 Solve the proportion $\frac{5}{8} = \frac{x}{40}$.

Cross-multiply: $8x = 200$, so $x = 25$.

- 3 A recipe requires flour and sugar in the ratio 3 : 2. If 12 cups of flour are used, how much sugar is needed?

$$\text{Sugar} = \frac{2}{3} \times 12 = 8 \text{ cups.}$$

- 4 If $a : b = 4 : 7$ and $b = 35$, find a .

$$\frac{a}{4} = \frac{35}{7} = 5, \text{ so } a = 20.$$

Unit rates

- 5 A car uses 8 gallons of gas to travel 224 miles. Find its fuel efficiency in miles per gallon.

$$\frac{224}{8} = 28 \text{ miles per gallon.}$$

- 6 A printer produces 150 pages in 6 minutes. Find its rate in pages per minute, and use it to find how long it takes to print 400 pages.

$$\text{Rate} = \frac{150}{6} = 25 \text{ pages per minute. Time} = \frac{400}{25} = 16 \text{ minutes.}$$

Percentage change and applications

- 7 A price increases from 60 to 75 dollars. Find the percentage increase.

$$\frac{75 - 60}{60} \times 100 = 25\%.$$

- 8 A jacket originally 120 dollars is discounted 30 percent. Find the sale price.

$$120 \times 0.7 = 84 \text{ dollars.}$$

- 9 A population decreases by 15 percent one year, then increases by 15 percent the next. Is the final population the same as the original?

Start with 100. After a 15% decrease: 85. After a 15% increase: $85 \times 1.15 = 97.75$. No — the final population is 2.25% less than the original.

- 10 An investment of 2000 dollars grows to 2300 dollars after one year. Find the percentage increase.

$$\frac{2300 - 2000}{2000} \times 100 = 15\%.$$

Weighted averages

- 11 A student scores 80 on a test worth 40 percent of the grade and 90 on a test worth 60 percent of the grade. Find the weighted average grade.

$$0.4(80) + 0.6(90) = 32 + 54 = 86.$$

- 12 A store sells 3 items at 10 dollars each and 7 items at 20 dollars each. Find the average price per item sold.

$$\text{Total revenue} = 3(10) + 7(20) = 30 + 140 = 170 \text{ dollars. Total items} = 10. \text{Average} = \frac{170}{10} = 17 \text{ dollars.}$$

Scale and similar figures

- 13 A model car is built at a scale of 1 : 24. If the actual car is 4.8 meters (480 cm) long, find the length of the model, in centimeters.

$$\frac{480}{24} = 20 \text{ centimeters.}$$

- 14 A map has a scale of 1 cm to 5 km. If two cities are 8.5 cm apart on the map, find the actual distance between them.

$$8.5 \times 5 = 42.5 \text{ km.}$$

Interpreting data tables

- 15 A table shows the hours worked by 4 employees in a week: A: 32, B: 40, C: 28, D: 36. Find the total hours worked and the mean hours per employee.

$$\text{Total} = 32 + 40 + 28 + 36 = 136. \text{Mean} = \frac{136}{4} = 34 \text{ hours.}$$

- 16 A table shows quarterly revenue, in thousands of dollars: Q1: 50, Q2: 65, Q3: 70, Q4: 55. Find the percentage of annual revenue earned in Q3, to the nearest tenth of a percent.

$$\text{Total} = 50 + 65 + 70 + 55 = 240. \text{Q3 percentage} = \frac{70}{240} \times 100 \approx 29.2\%.$$

Bar graphs and data displays

- 17 The bar graph shows a company's monthly sales, in thousands of dollars. Find the percentage increase in sales from January to March.

$$\frac{65 - 40}{40} \times 100 = 62.5\%.$$

Proportional reasoning in context

- 18 A factory produces 450 units using 6 machines running for 8 hours. Assuming the same rate per machine-hour, how many units would 9 machines produce in 8 hours?

$$\text{Rate} = \frac{450}{6 \times 8} = \frac{450}{48} = 9.375 \text{ units per machine-hour. New machine-hours} = 9 \times 8 = 72. \text{ Units} \\ = 72 \times 9.375 = 675.$$

- 19 If 5 workers can build a fence in 12 days, how many days would it take 4 workers, assuming the same work rate per worker?

$$\text{Total work} = 5 \times 12 = 60 \text{ worker-days. Days for 4 workers} = \frac{60}{4} = 15 \text{ days.}$$

A well-designed word problem: customer growth for a small business

- 20 This question has three parts. In the first quarter, a business had 240 customers, and 180 of them made a repeat purchase. (a) Find the repeat-purchase rate as a percentage. (b) In the second quarter, the number of customers grew by 25 percent, and the repeat-purchase rate held steady at the percentage found in part (a). Find the number of repeat customers in the second quarter. (c) The business sets a goal to raise the repeat-purchase rate to 80 percent by the third quarter, while keeping the same total number of customers as the second quarter. Find how many additional repeat customers (beyond part (b)'s total) are needed to reach this goal.

$$\text{(a) } \frac{180}{240} \times 100 = 75\%. \text{ (b) New customers} = 240 \times 1.25 = 300. \text{ Repeat customers} \\ = 0.75 \times 300 = 225. \text{ (c) } 80\% \text{ of } 300 \text{ is } 0.8 \times 300 = 240 \text{ repeat customers needed. Additional} \\ \text{customers needed} = 240 - 225 = 15.$$

Multiple choice — easy

21 Solve the proportion $\frac{14}{4} = \frac{x}{34}$ for x .

a. 121

b. 119 ✓

c. 117

d. 4

Cross-multiply: $4x = 14 \times 34 = 476$, so $x = 119$.

22 Solve the proportion $\frac{13}{13} = \frac{x}{52}$ for x .

a. 52 ✓

b. 13

c. 50

d. 54

Cross-multiply: $13x = 13 \times 52 = 676$, so $x = 52$.

23 Solve the proportion $\frac{4}{5} = \frac{x}{30}$ for x .

a. 26

b. 22

c. 24 ✓

d. 5

Cross-multiply: $5x = 4 \times 30 = 120$, so $x = 24$.

24 Solve the proportion $\frac{10}{9} = \frac{x}{18}$ for x .

a. 9

b. 20 ✓

c. 22

d. 18

Cross-multiply: $9x = 10 \times 18 = 180$, so $x = 20$.

25 Solve the proportion $\frac{3}{7} = \frac{x}{7}$ for x .

a. 1

b. 3 ✓

c. 7

d. 5

Cross-multiply: $7x = 3 \times 7 = 21$, so $x = 3$.

26 Solve the proportion $\frac{2}{13} = \frac{x}{26}$ for x .

a. 13

b. 4 ✓

c. 2

d. 6

Cross-multiply: $13x = 2 \times 26 = 52$, so $x = 4$.

27 Solve the proportion $\frac{10}{6} = \frac{x}{42}$ for x .

a. 70 ✓

b. 6

c. 72

d. 68

Cross-multiply: $6x = 10 \times 42 = 420$, so $x = 70$.

28 Solve the proportion $\frac{2}{3} = \frac{x}{12}$ for x .

a. 3

b. 8 ✓

c. 10

d. 6

Cross-multiply: $3x = 2 \times 12 = 24$, so $x = 8$.

29 Solve the proportion $\frac{10}{12} = \frac{x}{24}$ for x .

a. 18

b. 20 ✓

c. 22

d. 12

Cross-multiply: $12x = 10 \times 24 = 240$, so $x = 20$.

30 Solve the proportion $\frac{2}{8} = \frac{x}{28}$ for x .

a. 8

b. 9

c. 5

d. 7 ✓

Cross-multiply: $8x = 2 \times 28 = 56$, so $x = 7$.

31 Solve the proportion $\frac{14}{4} = \frac{x}{42}$ for x .

a. 4

b. 149

c. 147 ✓

d. 145

Cross-multiply: $4x = 14 \times 42 = 588$, so $x = 147$.

32 Solve the proportion $\frac{8}{8} = \frac{x}{29}$ for x .

a. 31

b. 8

c. 29 ✓

d. 27

Cross-multiply: $8x = 8 \times 29 = 232$, so $x = 29$.

33 Solve the proportion $\frac{8}{8} = \frac{x}{39}$ for x .

a. 37

b. 41

c. 8

d. 39 ✓

Cross-multiply: $8x = 8 \times 39 = 312$, so $x = 39$.

34 Solve the proportion $\frac{12}{3} = \frac{x}{46}$ for x .

a. 184 ✓

b. 186

c. 182

d. 3

Cross-multiply: $3x = 12 \times 46 = 552$, so $x = 184$.

35 Solve the proportion $\frac{11}{4} = \frac{x}{52}$ for x .

a. 141

b. 4

c. 143 ✓

d. 145

Cross-multiply: $4x = 11 \times 52 = 572$, so $x = 143$.

Multiple choice — medium

36 A budget decreases from 40 to 20. Find the percentage decrease.

a. 50% ✓

b. 55%

c. 100%

d. 45%

$$\frac{20}{40} \times 100 = 50\%.$$

37 A population increases from 220 to 264. Find the percentage increase.

a. 40%

b. 20% ✓

c. 25%

d. 15%

$$\frac{44}{220} \times 100 = 20\%.$$

38 A population increases from 140 to 147. Find the percentage increase.

- a. 10%
- b. 0%
- c. 8%
- d. 5% ✓

$$\frac{7}{140} \times 100 = 5\%.$$

39 An attendance figure increases from 240 to 312. Find the percentage increase.

- a. 30% ✓
- b. 25%
- c. 35%
- d. 60%

$$\frac{72}{240} \times 100 = 30\%.$$

40 A subscription count increases from 20 to 29. Find the percentage increase.

- a. 40%
- b. 45% ✓
- c. 90%
- d. 50%

$$\frac{9}{20} \times 100 = 45\%.$$

41 A membership total increases from 60 to 78. Find the percentage increase.

- a. 30% ✓
- b. 35%
- c. 25%
- d. 60%

$$\frac{18}{60} \times 100 = 30\%.$$

42 A population increases from 140 to 196. Find the percentage increase.

a. 35%

b. 40% ✓

c. 80%

d. 45%

$$\frac{56}{140} \times 100 = 40\%.$$

43 A membership total decreases from 60 to 30. Find the percentage decrease.

a. 55%

b. 100%

c. 45%

d. 50% ✓

$$\frac{30}{60} \times 100 = 50\%.$$

44 A population increases from 220 to 319. Find the percentage increase.

a. 50%

b. 40%

c. 45% ✓

d. 90%

$$\frac{99}{220} \times 100 = 45\%.$$

45 A budget decreases from 120 to 108. Find the percentage decrease.

a. 10% ✓

b. 5%

c. 15%

d. 20%

$$\frac{12}{120} \times 100 = 10\%.$$

46 An attendance figure increases from 160 to 232. Find the percentage increase.

a. 40%

b. 50%

c. 45% ✓

d. 90%

$$\frac{72}{160} \times 100 = 45\%.$$

47 A population decreases from 260 to 169. Find the percentage decrease.

a. 30%

b. 70%

c. 35% ✓

d. 40%

$$\frac{91}{260} \times 100 = 35\%.$$

48 A price increases from 260 to 377. Find the percentage increase.

a. 40%

b. 90%

c. 45% ✓

d. 50%

$$\frac{117}{260} \times 100 = 45\%.$$

49 An attendance figure decreases from 300 to 180. Find the percentage decrease.

- a. 80%
- b. 45%
- c. 35%
- d. 40% ✓

$$\frac{120}{300} \times 100 = 40\%.$$

50 An attendance figure decreases from 60 to 51. Find the percentage decrease.

- a. 10%
- b. 20%
- c. 30%
- d. 15% ✓

$$\frac{9}{60} \times 100 = 15\%.$$

Multiple choice — hard

51 A student scores 60 on an assignment worth 75% of the grade and 100 on an assignment worth 25% of the grade. Find the weighted average grade.

- a. 80
- b. 70 ✓
- c. 67
- d. 73

$$0.75(60) + 0.25(100) = 45 + 25 = 70.$$

- 52** A student scores 85 on an assignment worth 20% of the grade and 50 on an assignment worth 80% of the grade. Find the weighted average grade.

- a. 60
- b. 67
- c. 57 ✓
- d. 54

$$0.2(85) + 0.8(50) = 17 + 40 = 57.$$

- 53** A student scores 80 on an assignment worth 50% of the grade and 94 on an assignment worth 50% of the grade. Find the weighted average grade.

- a. 87 ✓
- b. 91
- c. 84
- d. 90

$$0.5(80) + 0.5(94) = 40 + 47 = 87.$$

- 54** A student scores 60 on an assignment worth 20% of the grade and 50 on an assignment worth 80% of the grade. Find the weighted average grade.

- a. 55
- b. 49
- c. 56
- d. 52 ✓

$$0.2(60) + 0.8(50) = 12 + 40 = 52.$$

- 55 A student scores 65 on an assignment worth 40% of the grade and 95 on an assignment worth 60% of the grade. Find the weighted average grade.

- a. 87
- b. 80
- c. 83 ✓
- d. 86

$$0.4(65) + 0.6(95) = 26 + 57 = 83.$$

- 56 If 9 workers can complete a task in 24 days, how many days would it take 8 workers, assuming the same work rate per worker?

- a. 29
- b. 27 ✓
- c. 24
- d. 25

$$\text{Total work} = 9 \times 24 = 216 \text{ worker-days. Days for 8 workers} = \frac{216}{8} = 27.$$

- 57 If 6 workers can complete a task in 11 days, how many days would it take 3 workers, assuming the same work rate per worker?

- a. 11
- b. 20
- c. 22 ✓
- d. 24

$$\text{Total work} = 6 \times 11 = 66 \text{ worker-days. Days for 3 workers} = \frac{66}{3} = 22.$$

58 If 3 workers can complete a task in 25 days, how many days would it take 5 workers, assuming the same work rate per worker?

- a. 25
- b. 17
- c. 15 ✓
- d. 13

Total work = $3 \times 25 = 75$ worker-days. Days for 5 workers = $\frac{75}{5} = 15$.

59 If 9 workers can complete a task in 12 days, how many days would it take 6 workers, assuming the same work rate per worker?

- a. 16
- b. 12
- c. 20
- d. 18 ✓

Total work = $9 \times 12 = 108$ worker-days. Days for 6 workers = $\frac{108}{6} = 18$.

60 If 8 workers can complete a task in 30 days, how many days would it take 10 workers, assuming the same work rate per worker?

- a. 24 ✓
- b. 22
- c. 26
- d. 30

Total work = $8 \times 30 = 240$ worker-days. Days for 10 workers = $\frac{240}{10} = 24$.

- 61** A map has a scale of 1 cm to 18 km. If two cities are 17 cm apart on the map, find the actual distance between them.

- a. 308
- b. 310
- c. 306 ✓
- d. 304

$$17 \times 18 = 306 \text{ km.}$$

- 62** A map has a scale of 1 cm to 17 km. If two cities are 15.5 cm apart on the map, find the actual distance between them.

- a. 267.5
- b. 265.5
- c. 261.5
- d. 263.5 ✓

$$15.5 \times 17 = 263.5 \text{ km.}$$

- 63** A map has a scale of 1 cm to 13 km. If two cities are 9.5 cm apart on the map, find the actual distance between them.

- a. 127.5
- b. 121.5
- c. 123.5 ✓
- d. 125.5

$$9.5 \times 13 = 123.5 \text{ km.}$$

- 64** A map has a scale of 1 cm to 11 km. If two cities are 6.5 cm apart on the map, find the actual distance between them.
- a. 69.5
 - b. 71.5 ✓
 - c. 73.5
 - d. 75.5

$$6.5 \times 11 = 71.5 \text{ km.}$$

- 65** A map has a scale of 1 cm to 5 km. If two cities are 13 cm apart on the map, find the actual distance between them.
- a. 67
 - b. 69
 - c. 65 ✓
 - d. 63

$$13 \times 5 = 65 \text{ km.}$$